SAIBALAJI NEELI

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EDUCATION

Oregon State University: M.S. Computer Science and Engineering |CGPA: 3.50 **Present**

Coursework: Machine Learning, Machine Learning Challenges, Topological Data Analysis, Human Computer Interaction, Algorithms, Operating systems, Error Correcting Codes.

ANNA UNIVERSITY: B.E. Computer Science and Engineering |CGPA: 3.60 **Apr 2023**

Coursework: Artificial Intelligence, Computer Architecture, Databases, Networks, Software engineering, Data warehouse.

WORK EXPERIENCE

TDK InvenSense – People counting using low-power MEMS time of flight sensor **Sep–Dec 2024**

- Developed a low-power people counting system using ultrasonic sensors, reducing energy consumption for scalable deployment.
- Designed and optimized ML models for embedded systems, ensuring accuracy and efficiency in resource-constrained environments.
- Integrated hardware and software solutions to create a functional prototype with seamless operation.

Super tutor AI – ML Development intern. New York | **Jun- Sep 2024**

- Developed machine learning models using Python, **TensorFlow**, and **scikit-learn** to enhance recommendation systems, increasing user engagement by 25% and retention rates by 15%.
- Collaborated with user feedback teams to identify key feature improvement areas, implementing changes that boosted app usability and resulted in a 50% rise in positive app store reviews.
- Analyzed user behavior by executing complex **SQL queries** on a dataset of over 500,000 records, uncovering insights that improved engagement metrics by 15%.
- Optimized **machine learning workflows** to maintain **high accuracy** despite data gaps.

Outlier – Prompt Engineer. San Francisco, California| **Apr-Jun 2024**

- Enhanced AI system functionality by debugging and optimizing response generation pipelines, increasing user satisfaction.
- Improved model accuracy and relevance by fine-tuning algorithms to better align outputs with organizational objectives.
- Collaborated with cross-functional teams to deploy AI solutions tailored to user needs, ensuring seamless integration and usability.

Oregon State University – Graduate Research Assistant. Corvallis, Oregon | **Jan-Mar2024**

- Conducted advanced research on minimal imputations in machine learning data preparation, leading to a 15% reduction in data processing time.
- Evaluated over 10 machine learning models to optimize data completeness while maintaining high predictive accuracy, improving overall efficiency by 25%.
- Co-authored a research paper for VLDB 2024, demonstrating expertise in data preprocessing, machine learning, and statistical analysis.

PROJECTS

Life Expectancy Prediction **Jan 2024**

- Developed a regression model to predict life expectancy using a dataset of 200,000+ records spanning 15 years (2000-2015).
- Achieved a high model performance with an R^2 value of 0.85.
- Preprocessed the dataset to handle missing values, outliers, and inconsistencies to enhance model accuracy.
- Utilized Python and scikit-learn for data analysis and model development.

Pneumonia Detection and Classification **Apr 2023**

- Implemented Convolutional Neural Networks (CNNs) for pneumonia detection using chest X-ray images.
- Achieved 95% classification accuracy and 0.92 ROC-AUC, surpassing the baseline models.
- Incorporated Transfer Learning with pre-trained models for improved diagnostic performance.
- Leveraged TensorFlow and Keras to streamline model training and testing processes.

Database Management System **Jan 2023**

- Designed and implemented a centralized SQL database to manage F-1 race data, consisting of over 10,000 race statistics.
- Optimized database queries, reducing data retrieval time by 30%.
- Applied indexing and normalization techniques to improve system performance.
- Utilized MySQL and SQL Server for database creation, management, and analysis.

SKILLS

- **Languages:** C, C++, Python, JavaScript, Java, HTML, CSS, R programming, SQL.
- **DB's:** MongoDB, MySQL, PostgreSQL | **Tools:** Microsoft Excel, Tableau, PowerBI, Jupyter, Google colab, Matplotlib.
- **Framework/Libraries:** Pandas, NumPy, TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, Hadoop| **OS:** Windows, Linux, MacOS.
- **Machine Learning**: Supervised and Unsupervised Learning, Neural Networks, Deep Learning, Reinforcement Learning, NLP.